Jessica U. Ezemba

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EDUCATION

Carnegie Mellon University — Pittsburgh, PA
Ph.D., Mechanical Engineering | GPA: 3.92/4.0 — Expected May 2027

Carnegie Mellon University — Pittsburgh, PA
M.S., Integrated Innovation for Products & Services | GPA: 3.96/4.0 — Dec. 2022

Texas A&M University — College Station, TX
B.S., Mechanical Engineering | GPA: 3.52/4.0 — May 2021

RESEARCH INTERESTS

I research at the intersection of artificial intelligence and engineering design. My work spans **vision-language models** for engineering simulations, **multi-agent orchestration systems**, **graph representation and learning**, and **neural surrogate modeling**. My future research aims to develop autonomous AI-driven design systems that integrate physical reasoning with foundation models to accelerate discovery across engineering domains.

PUBLICATIONS

Published

1. **Ezemba, J.,** Pohl, J., Tucker, C., & McComb, C. (2026). "A Large-Scale Benchmark to Assess Vision-Language Model Question Answering Capabilities in Engineering Simulations." Communications Engineering (Nature Portfolio). doi:10.1038/s44172-026-00786-2
   Project page: jessicaezemba.com/openseesime-page
2. **Ezemba, J.,** McComb, C., & Tucker, C. (2026). "Organizationally-Intelligent Agents: Connecting Multi-Agent Engineering Systems to Organizational Design Theory." Design Computing and Cognition '26, Springer Nature.
   Project page: jessicaezemba.com/oia
3. **Ezemba, J.,** Afful, J., & Wang, M. (2026). "Semantic-aware Implicit Neural Compression for Physics Simulations." PASC 2026. Peer-reviewed two-page summary, poster competition.
   Project page: jessicaezemba.com/SINCPS
4. **Ezemba, J.,** McComb, C., & Tucker, C. (2026). "Structure Shapes Coordination: An Empirical Evaluation of Organizational Design Principles in Multi-Agent Engineering Design." IDETC 2026.
5. Mehta, M.D.*, **Ezemba, J.*,** Tucker, C., & McComb, C. (2026). "VizArQA: A Benchmark for Visual Question Answering in Architectural Simulation." J. of Eng. for Sustainable Buildings and Cities.
6. **Ezemba, J.,** McComb, C., & Tucker, C. (2026). "Neural Network Surrogate Modeling for SFEM Using 3D Graph Representations: A Comparative Study." Journal of Mechanical Design, Vol. 148, No. 1.
7. **Ezemba, J.,** McComb, C., & Tucker, C. (2025). "Neural Network Surrogate Modeling for Stochastic FEM Using 3D Graph Representations: A Comparative Study." IDETC 2025.
8. **Ezemba, J.,** McComb, C., & Tucker, C. (2025). "Simulation vs. Hallucination: Assessing VLM QA Capabilities in Engineering Simulations." DESTION 2025, pp. 1-9.
9. **Ezemba, J.,** Afful, J., & Wang, M. (2025). "PhySiViT: A Physics Simulation Vision Transformer." SC25 Poster.
10. Tarutira, T.T., Alagbe, I., **Ezemba, J.,** Berhanu, E., & Ndashimye, E. (2024). "Baboon Detection for Smart Agriculture Using Deep Learning Algorithms." IEEE UV 2024.
11. **Ezemba, J.,** Cunningham, J.D., & Tucker, C.S. (2023). "Parameter Extraction From Images Using Multilabel Supervised Learning." IDETC 2023.

12. **Ezemba, J.,** & Layton, A. (2022). "Bio-Inspired Avenues for Advancing Brain Injury Prevention." Journal of Mechanical Design, Vol. 144, No. 12.
13. Khaled, M., Srinivasan, S., Toure, A., Chen, M., Kincaid, E., Lopaz, T., ..., **Ezemba, J.,** et al. (2021). "DREAMS: Drilling and Extraction Automated System." arXiv:2106.05874.

PHD RESEARCH

Carnegie Mellon University Pittsburgh, PA

PhD Candidate, Mechanical Engineering Aug. 2023 - Present

Advisors: Prof. Christopher McComb, Prof. Conrad Tucker

- Built custom Multi-Agent System (MAS) for engineering design and simulation utilizing organizational design principles, increasing computational efficiency by 50%
- Developed Model Context Protocol (MCP) server for prompt-to-CAD, parametric geometry variation, and FEA/CFD simulations, enabling faster integration into MAS and reducing computational costs by 50%
- Created comprehensive benchmark datasets (200,000+ questions, 10,000 simulations) for evaluating LLM performance on engineering simulation reasoning across text, image, and video modalities
- Engineered neural network surrogate models with FEniCSx to quantify uncertainty in complex engineering simulations, reducing computational costs by 90% across 16,000+ 3D geometries
- Advanced the application of Geometric Deep Learning to Graph Neural Networks, creating custom data loaders that effectively process complex 3D unstructured mesh geometries
- Integrated state-of-the-art LLMs and Vision Transformers for multimodal engineering data interpretation
- Architected custom self-retrieval and agentic RAG systems using LangGraph, automating simulation interpretation and reducing analysis time from days to hours

INDUSTRY COLLABORATIONS

Research partnerships developed during PhD

Siemens May 2026 - Present

Research Collaborator, Pre-Research and Development (PRD)

- Collaborating with the Siemens Pre-Research and Development organization on PhD research connecting foundation model agents to production CAD/CAE infrastructure
- Extending simulation-interpretation and agent-driven design research onto live Siemens NX and NX Nastran workflows using real aerospace components

Boeing Oct. 2025 - Present

Research Collaborator, AI for Engineering Design

- Partnered with Boeing researchers on multi-agent system design for engineering applications, resulting in two publications (IDETC 2026, DCC 2026)
- Applied organizational design theory to coordinate AI agent teams for complex engineering design and simulation tasks

Ansys Jan. 2025 - Present

Research Collaborator, Vision-Language Models for Simulation

- Collaborated with Ansys on developing large-scale benchmarks for evaluating vision-language model capabilities in engineering simulation environments
- Built OpenSeeSimE dataset (200,000+ questions, 10,000 simulations) and presented findings at DESTION 2025 with NSF sponsorship

SKILLS

Programming & Engineering: Python, FEniCSx, PyAnsys, ANSYS, Siemens NX, Simcenter 3D, NX Nastran, MATLAB, SOLIDWORKS, CATIA V5, AutoCAD, PTC Creo, LabVIEW
AI/ML: PyTorch, PyTorch Geometric, Multi-Agent Systems, LangGraph, Vision Transformers, GNNs, Geometric Deep Learning, Autoencoders, LLMs, RAG, MCP
Computation & DevOps: HPC, FEA, CFD, Probabilistic Simulation, Surrogate Modeling, Neural Operators, Linux, AWS
Engineering: Controller Design, Optimization Design, Product Design, Prototyping, GD&T

PROFESSIONAL EXPERIENCE

Siemens Remote / Princeton, NJ
PhD Research Intern, Pre-Research and Development (PRD) May 2026 - Aug. 2026

- Built an end-to-end CAD/CAE pipeline connecting AI agents to Siemens NX and Simcenter 3D through a 34-tool Model Context Protocol (MCP) server over NXOpen, covering feature-based geometry edits, meshing, NASTRAN solves (SOL 101/103/105), and result queries
- Designed ontology-guided engineering AI agents that reason from simulation observation to design action, extending the OODA loop with a Design step and a 157-category controlled vocabulary to produce auditable, machine-parseable inference traces
- Optimized three structural components against yielding, fatigue, buckling, and vibration requirements: the ontology agent achieved 2.5x the mass reduction of an unstructured baseline (9.3% vs. 3.7%) and met 2 of 3 coupled requirements at 91% of starting mass versus 1 of 3 at 123%
- Evaluated simulation interpretation across 17 AI models from 6 vendors on 550 structural FEA questions, with and without live solver tools, showing no configuration exceeds 67% accuracy

KLA Ann Arbor, MI
Advanced Mechatronics Intern May 2022 - Aug. 2022 & May 2023 - Aug. 2023

- Developed a 3D parametric geometry and physics-based optimizer to reduce the time to design flexures
- Created a 3D design exploration algorithm using ANSYS for fatigue, buckling, and frequency analysis
- Developed an optimization algorithm in MATLAB to decrease controller tuning time by over 50%
- Analyzed frequency response functions with loop shaping techniques to improve reliability of system
- Generated technical documents for the Frequency Response Analysis on MATLAB to train employees

Symbotic Wilmington, MA
Systems Design Intern Jan. 2023 - May 2023

- Increased warehouse automation throughput using FEA to optimize component mass and structural integrity of multiple 3D component parts
- Designed high-risk safety devices for warehouse staff, achieving 100% compliance with safety regulations
- Developed integrated documentation covering multiple subsystem aspects, improving overall system understanding

Spirit AeroSystems Wichita, KS
Research and Development Engineering Intern May 2021 - Aug. 2021

- Designed, prototyped, and introduced tooling aids to save over $100,000 in safety costs in Boeing programs
- Improved current method of skin defect detection on aircraft fuselage to reduce cost of rework
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Modeled engineering drawings of aircraft tooling using CATIA V5 to better document available models

ADDITIONAL RESEARCH EXPERIENCE

Carnegie Mellon University Pittsburgh, PA

Master Research Assistant Jan. 2022 - Aug. 2023

- Developed multilabel CNN model to extract over 2,000 design parameters from bicycle images, achieving 96% accuracy to enable automated reverse engineering of legacy designs
- Fine-tuned pre-trained ResNet50 model on grayscale engineering drawings to predict multi-type design parameters, reducing training time and computational costs compared to training from scratch

Texas A&M University College Station, TX

NASA Langley Research Design Project - Mechanical Design Lead June 2020 - May 2021

- Managed 6 engineers to build prototypes to meet NASA requirements and specifications
- Designed a drill rig base to meet specific constraints and criteria while on Mars with a $9,000 budget
- Selected as 1 of 10 universities to compete nationally in NASA Langley Research Center

Texas A&M University College Station, TX

Biomechanical Environments Laboratory - Undergraduate Research Assistant Aug. 2019 - May 2021

- Created CAD drawings for a pressure test chamber to help decrease death rate due to concussions
- Manufactured and assembled 7 prototype components in SolidWorks for pressure chamber
- Analyzed pressure data from 100 force impact tests to predict damage

Texas A&M University College Station, TX

Bio-inspired Sustainable Systems Laboratory - Undergraduate Research Assistant Oct. 2020 - May 2021

- Constructed framework for analyzing common protective mechanisms in nature from 8 organisms
- Published identified gaps and areas of improvement for protection using bioinspired methodologies

Art et Metiers Aix-en-Provence, France

Technical Partner May 2019 - June 2019

- Modeled an industrial mixer system using Finite Element Analysis Software, CATIA, to predict failure
- Identified failure in 3 shafts in an industrial mixer system to help reduce risk of injury

CONFERENCES / PRESENTATIONS

1. **Ezemba, J.,** McComb, C., & Tucker, C. "Do Human Organizational Principles Transfer to AI Agents? An Empirical Test Using Organizational Design Metrics." WE26, Society of Women Engineers, Boston, MA, 5-7 Nov. 2026. Research Presentation. (Upcoming)
2. **Ezemba, J.,** McComb, C., & Tucker, C. "Agent-Based Discovery of Simulation Interpretation Policies for Closed-Loop CAE." AESCAPE 2026 (USACM), Dallas, TX, 8-11 Sept. 2026. Research Presentation. (Upcoming)
3. **Ezemba, J.,** McComb, C., & Tucker, C. "Structure Shapes Coordination: An Empirical Evaluation of Organizational Design Principles in Multi-Agent Engineering Design." IDETC 2026, Houston, TX, Aug. 2026. Research Presentation.
4. **Ezemba, J.,** McComb, C., & Tucker, C. "Organizationally-Intelligent Agents: Connecting Multi-Agent Engineering Systems to Organizational Design Theory." Design Computing and Cognition '26, Ecole des Mines Paris, France, 8-10 July 2026. Research Presentation.
5. **Ezemba, J.,** Afful, J., & Wang, M. "Semantic-aware Implicit Neural Compression for Physics Simulations." PASC 2026, University of Bern & PHBern, Switzerland, 29 June - 1 July 2026. Poster Competition.

6. **Ezemba, J.,** McComb, C., & Tucker, C. "Do Human Organizational Principles Transfer to AI Agents? An Empirical Test Using Organizational Design Metrics." North East AI Agents Day 2026, New York, NY, 8 May 2026. Extended Abstract.
7. **Ezemba, J.,** Afful, J., & Wang, M. "PhySiViT: A Physics Simulation Vision Transformer." SC25, St. Louis, MO, 18 Nov. 2025. Poster Presentation.
8. **Ezemba, J.,** McComb, C., & Tucker, C. "Simulation vs. Hallucination: Assessing VLM QA Capabilities in Engineering Simulations." DESTION 2025, Irving, CA, 6 May 2025. NSF Sponsored Poster.
9. **Ezemba, J.,** McComb, C., & Tucker, C. "Neural Network Surrogate Modeling for Stochastic FEM using 3D Graph Representations." IDETC 2025, Anaheim, CA, 18 Aug. 2025. Research Presentation.
10. **Ezemba, J.,** McComb, C., & Tucker, C. "Parameter Extraction from Images Using Multilabel Supervised Learning." IDETC 2023, Boston, MA, 18 Aug. 2023. Research Presentation.
11. **Ezemba, J.** "Traumatic Brain Injury at the Cellular Level: Design of a Concussion Bioreactor." WE20 Student Research Competition, Virtual, 6 Nov. 2020. Conference Poster.
12. **Ezemba, J.** "Traumatic Brain Injury at the Cellular Level: Design of a Concussion Bioreactor." Gulf Coast Undergraduate Research Symposium, Virtual, 31 Oct. 2020. Research Poster.

TEACHING EXPERIENCE

Carnegie Mellon University Pittsburgh, PA

TA - Special Topics: Designing and Deploying AI/ML Systems (Graduate) Aug. 2025 - Dec. 2025

- Led full class sessions, held office hours, and created course content covering ethical AI/ML, automated machine learning, human-AI co-design, and agentic AI frameworks

Carnegie Mellon University Pittsburgh, PA

TA - Mechanics II: 3D Design (Undergraduate, ~150 students) Jan. 2025 - May 2025

- Led TA recitation sessions and office hours covering 3D statics, torsion, multiaxial stresses, failure criteria, and machine design

Texas A&M University College Station, TX

Lab Instructor - Experimental Physics and Engineering Lab II Jan. 2019 - May 2021

- Instructed a 3-hour mechanical physics weekly laboratory for 48 engineering students across 4 semesters

MENTORING

- Mentored an undergraduate researcher through CMU's Summer Undergraduate Research Apprenticeship (SURA) program in Summer 2026, advising work on scientific machine learning and quantum computing
- Co-advised one Integrated Innovation Institute masters student and one undergraduate MechE student with PI; each mentorship led to one journal publication
- Mentored 3 CMU Africa students; two won exchange program to CMU Pittsburgh; one secured position with EU tech company; acted as PI for Baboon Detection paper
- Mentored CMU ETIM masters student who secured internship and full-time offer at Phoenix manufacturing company

AWARDS AND HONORS

G. Sundback Graduate Fellowship in Mechanical Engineering, CMU	2026
NSF ACCESS PI Grant	2025
NSF Student Travel Award - DESTION 2025	2025
Best Poster Award - CAARMS	2024
DED BPart Committee Fellowship + Travel Scholarship - IDETC 2023	2023
Graduate Education for Minorities (GEM) Fellowship	2022

National Academy of Engineering (NAE) Grand Challenge Scholars	2021
Society of Women Engineers (SWE) Scholarship	2020
Benjamin Gilman Scholars Recipient	2019

SERVICE AND OUTREACH

- Human-Centered AI for K-12 Education (Moonshot Grant) - Assisted bringing 20 high school seniors from 4 PA districts to redesign K-12 education using human-centered AI with CMU faculty mentorship
- NAE Design Science & AI Workshop - Part of organizing team for official NAE-sponsored 3-day workshop in Kigali, Rwanda (2024)
- K-12 Outreach - Taught chemistry to high school students at Obama High School as part of CMU outreach
- MEGSO Mentorship Program Lead - Organized 3 industry speaker events and company visits for graduate students
- Conference Volunteering - CDFAM 2023 (NYC), IEEE Intl. Automated Vehicle Validation Conference 2024 (CMU)